

A 3D rendering of the text "DL for ENG" in a bold, gold, sans-serif font. The letters are floating on a deep blue ocean with visible ripples and small waves. The sky above is a mix of light blue and grey, with soft, white clouds. The overall scene has a cinematic, slightly surreal quality.

DL for ENG

# What You Will Be Able To Do

## THE GOALS, IN PLAIN WORDS

- **write** — the conservative form, and say why expanding it is wrong at an interface
- **recognise** — a pure-Neumann problem, and say what it needs to be well posed
- **scale** — a device problem before training, and say what breaks if you do not
- **hard-enforce** — a fixed-temperature face, and estimate what that buys
- **handle** — a curved boundary: the normal, the sampling, and the sign
- **identify** — an unknown conductivity from a few thermocouples

## WHAT TODAY ASSUMES

|           |                                            |
|-----------|--------------------------------------------|
| from L7.1 | residual, composite loss, hard enforcement |
| from L7.2 | Poisson, and its boundary conditions       |
| from L7.1 | the T3 quadrature machinery                |

## THE PHYSICS IS REAL NOW

Everything until here was a manufactured problem on a unit square. From this slide the units, the geometry and the material data are a device.

## AND ONE NEW CONDITION

Robin cannot be hard-enforced. It is the first condition that must stay soft.

# What Changes Now That the Physics Is Real

## CARRIED OVER UNCHANGED

- ✓ The network is the temperature field,  $T \approx \mathbb{N}\hat{w}(x, y)$ .
- ✓ Autograd supplies  $\nabla T$  and  $\nabla^2 T$  exactly.
- ✓ The loss is squared residuals, PDE plus conditions.
- ✓ Hard enforcement still beats penalties.
- ✓ Adam explores, L-BFGS converges.
- ✓ Sampling still obeys  $N_{\text{sample}} \geq P^*$ .

## NEW IN AN APPLICATION LECTURE

- Physical units — and therefore scaling problems.
- A third boundary condition: convection (Robin).
- Geometry that is not a rectangle.
- Material properties that may be unknown.
- An engineering question behind every solve.
- A tolerance that comes from the application.

**The engineering question behind L8.1:** given a device that dissipates a known power, how hot does it get and where? Everything in this lecture serves that question — which is also why we start with the steady state, the temperature a device settles at when it has been running long enough.

*L8.2 then asks how long it takes to get there, and what happens under a changing load.*

# From Fourier's Law to Poisson's Equation

## FOURIER'S LAW

$$\mathbf{q} = -k \nabla T$$

Heat flows down the temperature gradient, at a rate set by the conductivity  $k$ .

## ENERGY BALANCE

$$\nabla \cdot \mathbf{q} = Q \quad \Rightarrow \quad -\nabla \cdot (k \nabla T) = Q$$

In steady state, what flows out of a control volume equals what is generated inside it.

## CONSTANT $k$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{Q}{k} = 0$$

For a homogeneous material  $k$  comes out of the divergence, leaving Poisson's equation.

This is the same operator you met as Poisson in L7.2 — but  $Q/k$  now has units of  $K/m^2$ , and that changes how you must scale the problem before training.



# When the Conductivity Varies

## THE GENERAL FORM — $k$ INSIDE THE DIVERGENCE

$$\nabla \cdot (k(\mathbf{x}) \nabla T) + Q(\mathbf{x}) = 0$$

- **$k$  is rarely uniform** — aluminium bonded to ceramic, a pack of different cells
- **do not expand the divergence when  $k$  varies** — that form is wrong at an interface
- a PINN handles the conservative form directly, on the graph
- **across an interface  $k$  jumps and the gradient jumps with it** — the normal flux does not

## TWO MATERIALS, ONE INTERFACE



$T$  is continuous, its slope is not

# Three Thermal Boundary Conditions

## DIRICHLET — FIXED TEMPERATURE

$$T(\mathbf{x}) = \bar{T}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega_D$$

A surface held at a known temperature: a baseplate on a coolant channel, a contact with a large thermal mass.

## NEUMANN — PRESCRIBED FLUX

$$-k \frac{\partial T}{\partial n} = \bar{q}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega_N$$

A known heat flow per unit area. The special case  $\bar{q} = 0$  is an insulated or symmetric face, and is extremely common.

## ROBIN — CONVECTION

$$-k \frac{\partial T}{\partial n} = h(T - T_\infty)$$

The flux depends on the unknown temperature itself: Newton cooling into ambient air at  $T_\infty$  with film coefficient  $h$ .

*Robin is the one that is new relative to L7.2 — and it is the one that actually describes a device sitting in air. The next slide explains why it needs care.*

# Why the Convection Condition Is Harder

ROBIN RESIDUAL — BOTH A VALUE AND A DERIVATIVE OF THE UNKNOWN

$$\mathcal{L}_R = \frac{1}{n_R} \sum_{i=1}^{n_R} \left[ -k \frac{\partial T}{\partial n} - h(T - T_\infty) \right]^2$$

- Dirichlet fixes T, Neumann fixes the flux, Robin fixes a combination
- and Robin cannot be hard-enforced by a simple multiplier
- so it is a soft term by necessity, balanced against the residual
- the Biot number tells you how much that matters

BIOT NUMBER

$$Bi = \frac{hL}{k}$$

PRACTICAL GUIDANCE

Compute Bi before you train. If it is very large or very small, replace Robin with the limiting condition — you will get the same answer from a better-conditioned loss.

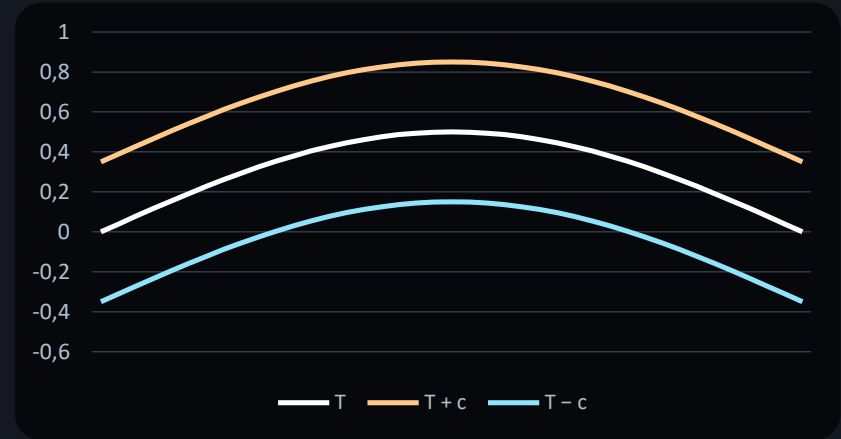
# Well-Posedness and the Pure-Neumann Trap

## COMPATIBILITY CONDITION FOR A PURE-NEUMANN PROBLEM

$$\oint_{\partial\Omega} \bar{q} \, ds = \int_{\Omega} Q \, d\Omega$$

- if every boundary carries a flux and none fixes a temperature,  $T$  is determined only up to a constant
- and a solution exists only if the fluxes balance the generation exactly
- **a PINN will not warn you** — it converges to some member of the family
- two fixes: pin the temperature at one point, or constrain the mean

## THE SAME FIELD, SHIFTED



all three satisfy the PDE and every flux condition



# Scaling: Do It Before You Train

CHANGE OF VARIABLES

$$\theta = \frac{T - T_0}{T_1 - T_0}, \quad \xi = \frac{x}{L}, \quad \eta = \frac{y}{L}$$

THE EQUATION KEEPS ITS FORM

$$\frac{\partial^2 \theta}{\partial \xi^2} + \frac{\partial^2 \theta}{\partial \eta^2} + \hat{Q} = 0$$

- a silicon device has **L ≈ 1 mm**, **k ≈ 150**, **Q ≈ 1e8** — raw numbers span many orders
- scale length by the device size and temperature by the physical rise
- every trial solution in this lecture assumes it
- **train scaled, present physical** — invert the transformation at the end

A SILICON POWER DEVICE

| QUANTITY         | PHYSICAL   | SCALED             |
|------------------|------------|--------------------|
| length L         | 1.0e-3 m   | <b>1</b>           |
| conductivity k   | 150 W/mK   | <b>1</b>           |
| source Q         | 1.0e8 W/m³ | <b>0.67</b>        |
| temperature rise | 0 → 45 K   | <b>0 → 1</b>       |
| film coeff. h    | 25 W/m²K   | <b>Bi = 1.7e-4</b> |

*Bi ≪ 1 here: the surface is nearly insulated, so the convection condition could be replaced by ∂T/∂n = 0 with negligible error.*

# Residual and Composite Loss

RESIDUAL

$$\mathcal{F}_{\hat{w}} = T_{,xx} + T_{,yy} + \frac{Q}{k}$$

COMPOSITE LOSS

$$\mathcal{L} = \mathcal{L}_{PDE} + w_D \mathcal{L}_{DBC} + w_N \mathcal{L}_{NBC}$$

- both second derivatives come from two autograd passes on one graph
- interior points test the residual; boundary points carry the soft conditions
- hard-enforce the Dirichlet term and it disappears from the loss entirely
- **the source is known in a forward problem** — make it unknown and this becomes inverse

|                     |                          |                                 |                     |                         |                                    |
|---------------------|--------------------------|---------------------------------|---------------------|-------------------------|------------------------------------|
| $\mathcal{L}_p^d$   | interior points          | <i>always present</i>           | $\mathcal{L}^{DBc}$ | fixed-temperature faces | <i>removable by trial solution</i> |
| $\mathcal{L}^{NBc}$ | flux and insulated faces | <i>soft; may need weighting</i> | $\mathcal{L}^R$     | convective faces        | <i>soft by necessity</i>           |

# Hard-Enforcing a Fixed-Temperature Face

## TRIAL SOLUTION FOR A PRESCRIBED TEMPERATURE ON ALL FOUR EDGES

$$T_{trial}(x, y) = \bar{T} + x(1 - x) y(1 - y) \mathcal{N}_{\hat{w}}(x, y)$$

- the same construction as L7.1, now reading as physics
- the network supplies the interior; the multiplier holds the face
- for a non-uniform prescribed temperature, evaluate it on the boundary
- **measured in L7.1** — about 165× more accurate in relative L2
- and the held face is usually the interface you care about

## WHICH FACES CAN BE HARD-ENFORCED

|                   |                  |
|-------------------|------------------|
| Fixed temperature | yes — multiplier |
|-------------------|------------------|

|                          |           |
|--------------------------|-----------|
| Insulated, $\bar{q} = 0$ | sometimes |
|--------------------------|-----------|

|                 |    |
|-----------------|----|
| Prescribed flux | no |
|-----------------|----|

|                    |    |
|--------------------|----|
| Convection (Robin) | no |
|--------------------|----|

*A mixed problem therefore always retains at least one soft term. Hard enforcement reduces the balancing problem; it rarely eliminates it.*

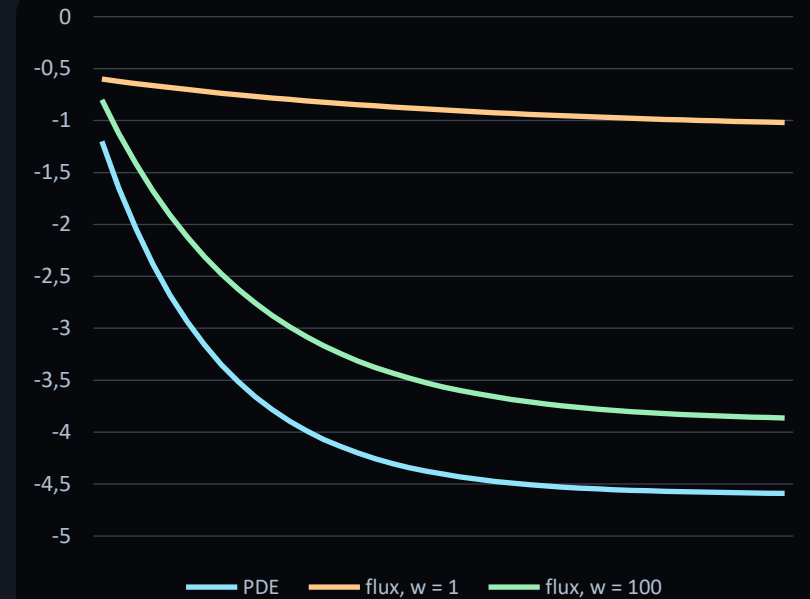
# Weighting the Flux Terms

## LIU'S DOCUMENTED EXCEPTION

$$w_N = 100$$

- Liu's default is one for every term, tuned only when a term is ignored
- **the reason is dimensional** — a flux residual and a PDE residual carry different units
- non-dimensionalising first removes most of it
- **diagnose before you tune** — plot the per-term losses separately

## PER-TERM LOSS HISTORY



*$\log_{10}$  loss. The flat amber curve is the failure: the total is falling, but the flux condition is not being learned.*

# Sizing the Network and the Sample Set

## PSEUDO-DIMENSION

$$P^* = (p + 1) + (N_n + 1) N_L$$

- $P^*$  counts neurons, not weights
- sample density should follow the temperature gradient, not the geometry
- concentrate points near sources, cooled faces and geometric features
- **a starting point** — ten to thirty times  $P^*$  interior points

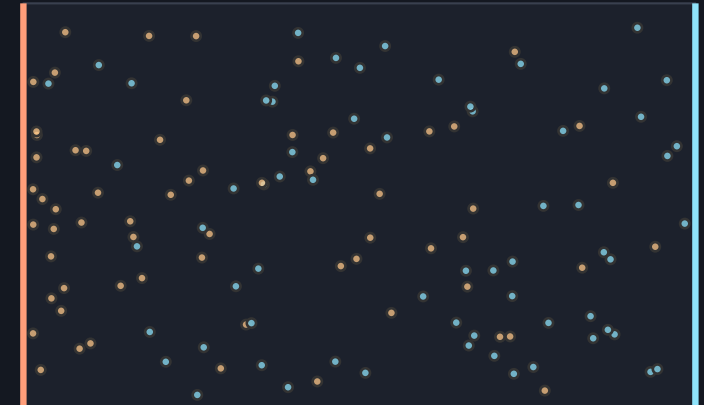
## SAMPLING CONDITION

$$N_{sample} \geq P^* \quad N_{sample} \gg P^*$$

### DENSITY FOLLOWS THE GRADIENT

heat source

cooled face



denser sampling where the gradient is steep

# Why Rectangles Are Not Enough

- every hard-enforcement trick so far has assumed a rectangle
- real thermal hardware is not rectangular
- **and geometry is not cosmetic** — a hole beside a source concentrates flux
- two things break: the multiplier has no closed form, and uniform sampling misses the feature

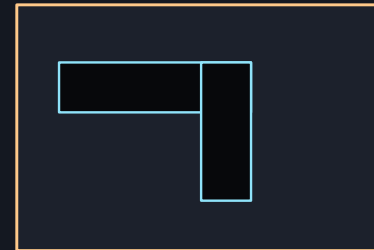
## GEOMETRIES THAT ACTUALLY OCCUR



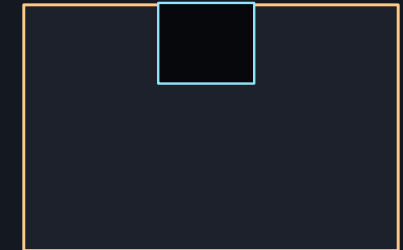
plate with a hole



fin array



channel with a bend



tab / notch



# Triangulating the Domain

## ELEMENT AREA

$$A = \frac{1}{2} |\mathbf{u} \times \mathbf{v}| = \frac{1}{2} |u_x v_y - u_y v_x|$$

## NODAL WEIGHT

$$w_i = \sum_{T \ni i} \frac{A_T}{3}$$

## QUADRATURE

$$\int_{\Omega} g \, d\Omega \approx \sum_{i=1}^n w_i g(\mathbf{x}_i), \quad \sum_{i=1}^n w_i = \int_{\Omega} 1 \, d\Omega$$

- **the mesh does not hold the solution** — the network still does
- it exists only to place sample points and supply integration weights
- a genuine difference from FEM: no shape functions, no assembly
- **and a coarse mesh suffices** — its job is coverage, not accuracy

## T3 MESH AROUND A HOLE



elements shrink towards the hole, where the flux concentrates

# A Learned Boundary Multiplier

## HARD ENFORCEMENT WITHOUT A CLOSED-FORM MULTIPLIER

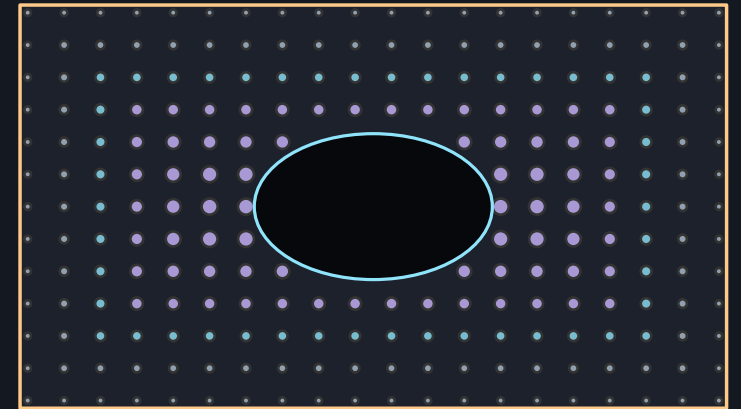
$$u_{trial}(x, y) = \mathcal{N}_{\hat{v}}^{DBC}(x, y) \mathcal{N}_{\hat{w}}(x, y)$$

- on a rectangle the multiplier is a product of linear factors
- on a plate with an elliptical hole there is no such closed form
- so train a small second network to be that function, once, in advance
- the cost is one cheap pre-training run
- for an analytic shape the level set itself will do

## ANALYTIC SHORTCUT FOR AN ELLIPSE

$$\left(\frac{x - x_c}{a}\right)^2 + \left(\frac{y - y_c}{b}\right)^2 = 1$$

## THE MULTIPLIER FIELD



zero on the outer edge, positive inside

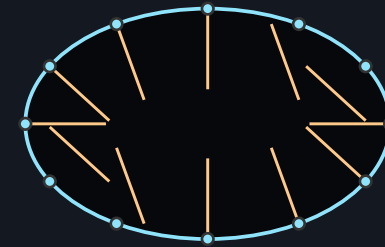
# Flux Conditions on a Curved Boundary

THE NORMAL DERIVATIVE, EVALUATED POINTWISE

$$\frac{\partial T}{\partial n} = \nabla T \cdot \mathbf{n}$$

- on a rectangle the normal is one of four constant vectors
- on a curved boundary it varies point to point
- **the recipe is unchanged** — autograd gives the gradient, you supply the normal
- get the sign right: the outward normal of the domain points into the hole
- and sample the curve by arc length, not by angle

NORMALS AROUND A HOLE



$n$  points OUT of the material, into the hole  
each sample point carries its own normal

# The Energy Route for Steady Conduction

TOTAL POTENTIAL FOR STEADY CONDUCTION

$$\Pi(T) = \frac{1}{2} \int_{\Omega} k \nabla T \cdot \nabla T \, d\Omega - \int_{\Omega} Q \, T \, d\Omega - \int_{\partial\Omega_N} \bar{q} \, T \, ds$$

- the steady field minimises this functional
- so the flux condition appears as a boundary work term and needs no loss
- in exchange the fixed-temperature condition becomes essential
- Liu's measured trade** — about 10× faster, typically less accurate

COLLOCATION OR ENERGY?

|                   | COLLOCATION | ENERGY       |
|-------------------|-------------|--------------|
| derivative order  | 2nd         | 1st          |
| flux condition    | loss term   | natural      |
| fixed temperature | either      | must be hard |
| needs quadrature  | no          | yes          |
| convergence       | slower      | ~10× faster  |
| accuracy          | higher      | lower        |

# Manufacturing a Solution to Check Against

CHOOSE THE ANSWER FIRST, THEN DERIVE THE SOURCE THAT PRODUCES IT

$$T = \sin(\pi x)\sin(\pi y) \quad \Rightarrow \quad \frac{Q}{k} = 2\pi^2 \sin(\pi x)\sin(\pi y)$$

- pick a field, substitute it in, and read off the source that makes it exact
- now you have a reference to check against
- this is the only way to separate discretisation error from formulation error
- do it before touching real geometry or real material data

## PRIMARY METRIC

$$\varepsilon_{L_2} = \frac{\|T_{\hat{w}} - T_{exact}\|_2}{\|T_{exact}\|_2}$$

## REPORT ALL THREE

Relative  $L_2$  for the overall fit.

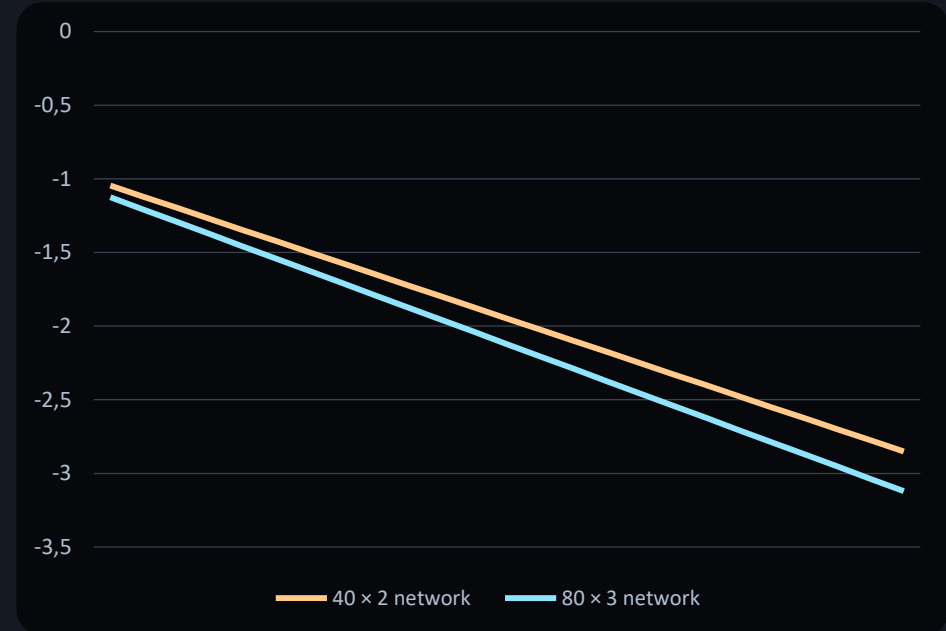
Maximum absolute error — the hot spot is what a thermal design is certified on.

Maximum error on each boundary, separately.

# Convergence: What Should Fall, and What Should Not

- a PINN has no mesh to refine, so the convergence study is different
- refine the sample set and the error should fall, then plateau
- the plateau tells you the network, not the sampling, is now the limit
- widening raises  $P^*$ , so the minimum sample count rises with it
- if nothing improves along any axis, the formulation is wrong

ERROR AGAINST SAMPLE COUNT

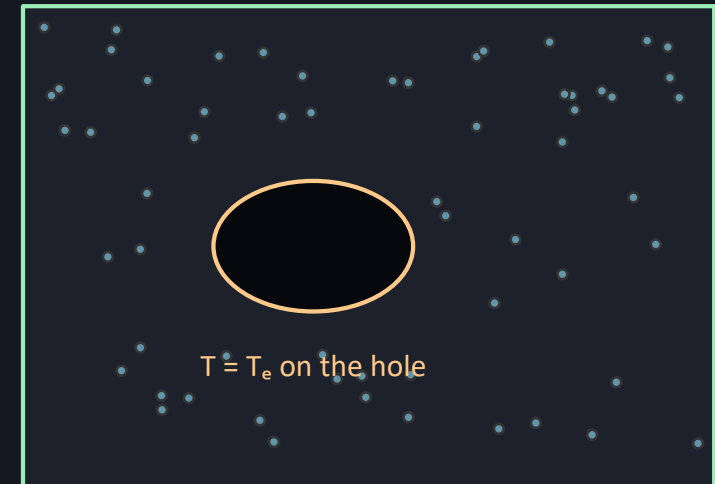


*$\log_{10}$  relative  $L_2$  against interior point count. Each curve flattens where the network, not the sampling, becomes the limit.*

# Heated Plate with a Cooling Hole

- the canonical device problem, and the one Liu carries through both chapters
- a plate generating heat, cooled through a hole
- fixed temperature on the hole, insulated outer edges, uniform generation
- the engineering question is the peak temperature and where it sits
- and the same geometry returns in L8.2 with time added

## PROBLEM DEFINITION



insulated outer edges ( $\bar{q} = 0$ )

uniform internal generation  $Q$  throughout

the peak sits on the edge furthest from the hole

# Building the Model, Step by Step

## 1 Scale

Map the plate to the unit square and the temperature rise to  $[0, 1]$ . Compute  $B_i$  to check whether convection matters.

## 2 Mesh

Triangulate the domain with elements refined towards the hole. Extract nodes and nodal weights.

## 3 Multiplier

Use the hole's level-set function to hard-enforce its fixed temperature. No pre-training needed for an ellipse.

## 4 Sample

Interior points from the mesh; boundary points by arc length on the hole and uniformly on the outer edges.

## 5 Assemble

Residual from two autograd passes; add the flux loss on the insulated edges with its own weight.

## 6 Train

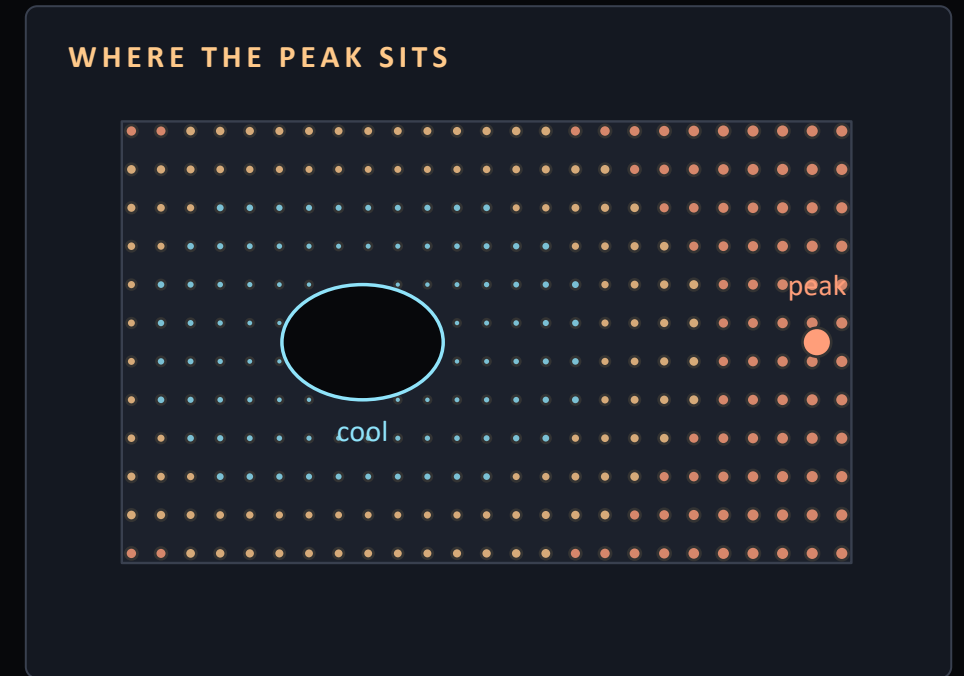
Adam to explore, L-BFGS to converge. Watch the per-term losses, not just the total.

Steps 1 and 2 are where most projects go wrong, and neither involves a neural network. Scaling and geometry come before any training.



# Reading the Result Like an Engineer

- a **contour plot is not a result** — a number with a tolerance is
- check the flux balance: what leaves through the hole must equal what is generated
- look at the residual map, not only the loss
- a small mean residual can hide a large local one exactly where it matters
- then ask the design question the model was built to answer



● **Peak temperature** against the limit

● **Residual map** no local hot spots

● **Flux balance** in = out, to a few %

● **Boundary error** on each face separately

# Identifying an Unknown Conductivity

THE FORWARD MODEL, WITH  $k$  PROMOTED TO A TRAINABLE PARAMETER

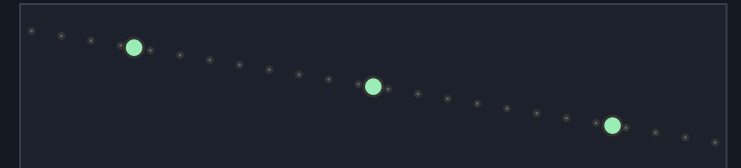
$$k \text{ trainable}, \quad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

- thermal conductivity is one of the least reliable numbers in a datasheet
- promote  $k$  to a trainable parameter and add a data term from a few thermocouples
- the same network, the same loss, one more trainable quantity
- this is the highest-value use of a PINN in thermal work
- **but watch identifiability** — sensors on an isotherm recover nothing

## SENSOR PLACEMENT MATTERS



clustered —  $k$  poorly identified



spread across the gradient —  $k$  recovered

# PINN or FEM for a Steady Thermal Problem?

## USE FEM

A single forward solve on fixed geometry with known properties.  
Sharp material interfaces.  
You need a certified answer with an established error bound.  
Decades of validation behind it.

## USE A PINN

Unknown parameters to identify from sparse sensors.  
Many solves across a parameter sweep, where a trained model evaluates instantly.  
Geometry that resists meshing.

- ▶ For one steady conduction solve on a meshable domain, FEM is faster, more accurate and better validated. Saying so is not a weakness of the course — it is what lets you recognise the cases where a PINN genuinely wins.
- ▶ Those cases are real and they are the ones this course targets: inverse problems, parametric studies, and coupling with data. All three are awkward for a classical solver and natural here.

*The honest position: a PINN is not a better solver. It is a differentiable one — and differentiability is what buys you inverse problems and design sensitivity.*

# Pitfalls Specific to Stationary Heat

**Unscaled physical units**

Conductivities of  $10^2$  and sources of  $10^8$  fed straight into a tanh network.

Symptom: loss barely moves from its initial value. Fix: non-dimensionalise first.

**Pure-Neumann degeneracy**

Every face carries a flux condition, so the temperature level is undetermined.

Symptom: perfect residual, wrong absolute temperature. Fix: pin one point.

**Flux term ignored**

The PDE residual dominates and the insulated faces are never learned.

Symptom: total loss falls, flux term flat. Fix: weight it, as Liu does with  $w^N = 100$ .

**Normal sign error**

The outward normal on an internal hole points into the hole, not away from it.

Symptom: heat flowing the wrong way; a plausible but inverted field.

**Material interface**

A smooth network cannot represent the kink in  $\nabla T$  at a conductivity jump.

Symptom: smeared interface. Fix: separate networks per material, coupled at the interface.

*The first two are formulation errors and no amount of training fixes them. Check both before you blame the optimiser.*

# Questions

## TEN TO TEST WHETHER TODAY LANDED

- why must the divergence stay unexpanded when  $k$  varies?
- what does a jump in  $k$  do to the gradient, and what stays continuous?
- why can a Robin condition not be hard-enforced?
- what is wrong with a problem whose boundaries are all flux conditions?
- **why scale before training** — and which construction depends on it?
- what does hard enforcement buy on a fixed-temperature face?

## AND FOUR MORE

- |              |                                                           |
|--------------|-----------------------------------------------------------|
| <b>seven</b> | why sample a curved boundary by arc length?               |
| <b>eight</b> | what does the triangulation do, if not hold the solution? |
| <b>nine</b>  | how do you check a steady result without a reference?     |
| <b>ten</b>   | when would you use FEM instead, and say so plainly?       |

## THE ONE WITH NO SETTLED ANSWER

Your sensors all sit on one isotherm. Can  $k$  be recovered at all?

## BEFORE L8.2

Ex\_08.1 notebooks 01 to 03. Bring the flux-balance number.

## Ex\_8.1 — Steady Conduction in a Plate with a Hole

### TASKS

- 1 Verify on a manufactured solution: solve on the unit square and reproduce the Ex\_7.1 accuracy.
- 2 Add an internal heat source with physical units, then non-dimensionalise and confirm the answers agree.
- 3 Introduce an elliptical hole; sample the domain and hard-enforce its fixed temperature by the level-set multiplier.
- 4 Impose zero flux on the outer edges. Sweep  $w^N$  over 1, 10, 100 and plot the per-term losses.
- 5 Report peak temperature, its location, and the flux balance in versus out.

### QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why must the problem be scaled before training? sl. 8

---

Why can a convection condition not be hard-enforced? sl. 6, 10

---

What breaks if every face carries a flux condition? sl. 7

---

How do you know the flux condition was actually learned? sl. 11, 22

---

When would you use FEM instead, and why? sl. 24

*Built on the same core library as Ex\_7.1 and Ex\_7.2 — only the residual, the conditions and the geometry change.*

# What Time Will Add in L8.2

|                      | L8.1 — STATIONARY      | L8.2 — DYNAMIC                                            |
|----------------------|------------------------|-----------------------------------------------------------|
| The equation         | $\nabla^2 T + Q/k = 0$ | $\partial T / \partial t = c \nabla^2 T + Q / (\rho c_p)$ |
| Inputs               | $x, y$                 | $x, y, t$                                                 |
| Conditions           | <i>BCs only</i>        | <i>BCs + one initial condition</i>                        |
| Domain               | $\Omega$               | $\Omega \times (0, t_{en}^d]$                             |
| New failure mode     | —                      | <i>causality: late times fit before early ones</i>        |
| Engineering question | <i>how hot?</i>        | <i>how fast, and under what duty cycle?</i>               |

Same geometry, same material, same code. One term is added to the operator, and one condition to the loss.

# Key Takeaways

- 1 It is Poisson, with units**

Steady conduction is the elliptic problem of L7.2 carrying physical quantities. The mathematics is unchanged; the scaling, the conditions and the geometry are not.
- 2 Scale before you train**

A device with  $L = 10^{-3}$  m and  $Q = 10^8$  W/m<sup>3</sup> will not train unscaled. Non-dimensionalisation is a necessary step, not a tidiness measure.
- 3 Convection resists hard enforcement**

Robin constrains a value and a derivative at once, so no multiplier satisfies it. It stays soft, and therefore needs weighting.
- 4 Geometry is the real work**

Meshing, multipliers and normals take more effort than the PINN itself. The mesh places points and supplies weights — it never holds the solution.
- 5 The win is the inverse problem**

For one forward solve, use FEM. For an unknown conductivity recovered from three thermocouples, use this.

*Next — L8.2: Dynamic Heat Transfer. Same plate, same hole, with time.*